Programming for Data Science
(CC4154)

1st Term | Finals
AY 2025-2026

Small end-to-end Data Science Python Application

2DSA2
Group 3

Caliba, Benedict S.
Veloso, Enrico Miguel
Yutuc, Thom Daniel C.

Abstract

In alignment to the Sustainable Development Goal 3 (Good Health and Well-being), this project aims to detect regions in the Philippines with low and high risks of outbreaks with regards to infectious diseases through predictive analysis. Using Logistic Regression for the baseline model and Random Forest Classifier for the advanced model, the researchers wanted to know what regions are more likely to have increased in the number of cases as time passes by. Models showing regions that have high population density are classified as ‘high risk’ that includes National Capital Region (NCR), Region IV-A (CALABARZON), Region III (Central Luzon), and Region XI (Davao Region). Moreover, regions that have low population density are classified as ‘low risk’ that includes NIR (Negros Island Region), Region I (Ilocos Region), and Region VIII (Eastern Visayas). However, the logistic regression did not yield any satisfactory results, whereas the random forest classifier displayed satisfactory results due to the quality of the dataset.

Introduction

The Philippines continues to face a high burden of infectious diseases such as dengue, tuberculosis, and leptospirosis, intensified rapid urbanization, frequent typhoons, and weak sanitation systems (DOH, 2023; WHO, 2022). These risks are concentrated in low income communities where overcrowding and limited access to clean water increase disease transmission, reflected in over 400,000 dengue cases reported in 2022 (PSA, 2023). Ongoing outbreaks and the country’s high global ranking in TB incidence highlight persistent regional disparities (World Bank, 2024). These challenges hinder toward Sustainable Development Goal 3, which emphasizes disease reduction and stronger early warning systems by 2030 (United Nations, 2015). Given widespread poverty and climate related risks, accurate outbreak risk classification for directing resources to high risk regions and strengthening health system resilience (UNDP, 2023). This study examines which regions in the Philippines face high or low outbreak risk using machine learning classification models to support targeted public health interventions.

Data and Methods

Data Source

The researchers extracted data from the official Department of Health (DOH) website under the EB Weekly Disease Surveillance Reports. It contains weekly data on how many cases were there in a barangay for each disease. Only available data were extracted as other datasets were inaccessible to the public. Thus, the dataset spans from January 20, 2025 to August 23, 2025.

Dataset

The dataset contains the following:

- **Regions:** All regions in the Philippines
- **PHI:** (Province / Highly Independent City)
- **Barangays:** Barangays that have reports of clusters
- **Diseases:** Common diseases reported
- **No. of Cases:** Number of cases for the previous four weeks

Preprocessing

Before handling the dataset, it must be cleaned to remove errors, duplicates, and inconsistencies in making insights with the help of a Python library *Pandas*. The following are done on each variable:

- **Regions:**
  - The Roman Numeral or the acronyms of the regions will be included in the column, and they will be in uppercase letters.
  - All numbers will be converted (e.g. 3 → III and 9 → IX)
  - REGION and the name of the region will be removed (e.g. REGION XI (DAVAO REGION → XI and REGION VI (WESTERN VISAYAS) → VI)
  - 4A and 4B will become IV-A and IV-B respectively.
- **PHI (Province / Highly Independent City):**
  - All values will also be in uppercase letters.
  - CITY OF will be removed and CITY will be appended after the city name (e.g. CITY OF MANILA → MANILA CITY)
  - Trailing whitespaces will be removed (e.g. DAVAO CITY → DAVAO CITY)
  - Ã‘ and Ã± will be replaced by Ñ.
  - LAS PIÑAS → LAS PIÑAS CITY
- **Barangays:**
  - All values will be in uppercase letters.
  - Trailing whitespaces will be removed
  - BARANGAY will be shortened to BRGY. and BRGY will add a period, producing BRGY.
  - Ã‘ and Ã± will be replaced by Ñ.
  - POB." will be removed.
  - Periods shall have space after the next text (e.g. R.A.PADILLA → R.A. PADILLA)
- **No. of Cases:**
  - The values are converted into numeric form as the feature will be part of the input features of the two models that will be utilized.

EDA Summary

During the exploration, there were 3 rows that have null values. These were dropped or removed from the dataset as it will leave errors and be irrelevant to the research. The dataset shows that in NCR, it has the most number of cases, regardless of diseases, in the whole country, and Region V has the least number of cases. Moreover, dengue has the most number of cases with a value of 79.

Model

The Logistic Regression Model is used as the baseline model to classify Philippine regions as high or low risk of an outbreak using key variables such as the number of cases per disease and also the morbidity week. This choice is supported by Carvajal et al (2018), who demonstrated that logistic regression is an effective baseline model for predicting dengue incidence in Metropolitan Manila using environmental factors such as rainfall, and humidity. On the other hand, Random Forest Classifier Model is used as the advanced model to improve prediction accuracy while maintaining the same risk categories. This selection is supported by Song et al. (2021), who showed that the Random Forest Model achieved the best prediction performance among several machine learning models for categorical predictions.

Application / User Interface:

The User Interface used the Python module Streamlit to view the dataset, predictions, and plots. In the Dataset page, the whole dataset is displayed the Region, PHI, Barangay, Disease, Number of Cases, and Morbidity Week. At the bottom, the user may add or more data if there are more cases reported. Meanwhile, in the Predictions page, basic information of the dataset will be shown that includes the number of records, average cases per location, and the number of high and low risks locations. Below the information is the model training setup where the user may choose between the Logistic Regression and Random Forest training models. Scrolling downwards, different data visualizations can be seen that compare between the models regarding their performance. Data exploration of the dataset is also displayed here.

Results

Key Findings:

The outbreak risk prediction system, which uses two machine learning models such as Logistics Regression Model and Random Forest Classifier Model yielded the following finding/s:

- **Superior Performance of RFCM** - The Random Forest Classifier Model demonstrated near-perfect performance, significantly outperforming the simpler Logistic Regression Model (LRM) across all evaluation metrics.

The system defined a region as "at risk" if it is predicted to have above the average number of cases (4.7 cases) in the following week. Based on the output of the high-performing Random Forest Classifier Model, the following is observed:

- The most frequently predicted high-risk regions are those with high population density and known disease incidence. Based on the classification logs from the high-performing model:
    - The regions observed to have a high frequency of "HIGH" risk predictions (i.e., multiple instances of above-average cases predicted for the following week) in the model output logs include: *National Capital Region (NCR), Region IV-A , (CALABARZON). Region III (Central Luzon), Region XI (Davao Region).*
    - Regions observed to have a lower frequency of "HIGH" risk predictions (i.e., more instances of "LOW" risk predicted for the following week) in the model output logs include: *NIR (Negros Island Region), Region I (Ilocos Region), Region VIII (Eastern Visayas)*

Model Performance:

The model performance comparison highlights the necessity of using an advanced, non-linear classifier for this prediction task. The system defined risk as a binary classification: 1 for high risk (next week's cases > average cases of 4.7) and 0 for low risk.

Metrics	Logistic Regression Model	Random Forest Classifier Model
Accuracy	63.86%	99.91%
Precision	43.30%	100.00%
Recall	52.48%	99.70%
F1 Score	47.45%	99.85%

The RFCM's near-100% scores in Precision, Recall, and F1 Score indicate that it is highly reliable, with an almost negligible rate of false alarms (False Positives) or missed outbreaks (False Negatives)

Patterns and Features:

Patterns in High-Risk Regions (NCR, IV-A, III, XI):

- The success of the RFCM in accurately classifying these regions as high-risk is driven by the strong influence of the categorical features (Region, PHI, Barangay, and Disease) which the majority are considered as an important feature.

- **Geographic Context Over Case Count:** The RFCM learned that the high-risk status is not solely dependent on the current number of cases, but is inherently tied to the location. Predictions for high-risk regions exhibit a pattern where risk is flagged:
    - When the current No. of Cases is above the overall average (4.7 cases).
    - Even for moderate case numbers, the model is biased toward a "HIGH" risk prediction because the Region feature signals a historical or environmental susceptibility (e.g., high population density or mobility) that makes future outbreaks more likely.
- **Pattern of True Positives:** The RFCM almost perfectly identifies a true "HIGH" risk scenario. This indicates a strong pattern recognition of localized conditions: a specific disease (e.g., Measles, Chikungunya) in a high-risk region (e.g., NCR) during a susceptible time (MW) almost always leads to a high-risk prediction for the following week.

Patterns in Low-Risk Regions (NIR, I, VIII):

- The prediction patterns in low risk regions show that the Random Forest Classifier Model reduces the influence of random numerical fluctuations by giving greater importance to location. This allows the model to make more stable and accurate low risk classifications despite variability in case counts.
    - **Location as a Stabilizing Feature:** For low-risk regions, the Region feature acts as a powerful dampener, preventing false alarms. Predictions for these regions consistently follow a pattern of "LOW" risk, even when the No. of Cases momentarily approaches or exceeds the average case threshold of 4.7.
    - **Pattern of True Negatives:** The RFCM is highly accurate at predicting a true "LOW" risk. This suggests these regions have intrinsic, identifiable characteristics (e.g., lower population density, successful local health interventions) that consistently prevent case counts from rising significantly in the next week, a pattern the RFCM successfully encodes via the one-hot encoded geographic features.

Model	Prediction Pattern in Relation to Regions	Implication
LRM	Inconsistent and Error-Prone. It frequently produces False Positives in low-risk regions (predicting HIGH when actual is LOW) and False Negatives in high-risk regions (predicting LOW when actual is HIGH) .	The risk pattern is viewed as a simple numerical relationship between current case count and week (No. of Cases and MW), ignoring the crucial factor of location.

RFCM	Highly Consistent and Localized. Predictions are strongly correlated with the known risk status of the region, only predicting "HIGH" risk when the location and disease context justify it, regardless of minor fluctuations in MW or No. of Cases	The risk pattern is viewed as a complex, non-linear function of location (Region, PHI, Barangay) and disease type, making it a robust and reliable predictor of localized outbreaks.

Discussion

In the context of SDG 3 Health and Well-being, the model helps understand potential outbreak locations. A limiting factor could be the quality of data used for the project. Future work will focus on improving the quality of collected data, as the model's reliability and performance are highly dependent on the data quality and accuracy. Although the model is functional, poor-quality data may lead to inaccurate or biased results.

Conclusion

In terms of programming, data collection, and exploratory data analysis, Caliba was responsible. Veloso created machine learning models and graphs, and Yutuc developed the front-end. Regarding the manuscript, the abstract, data, and methods were prepared by Caliba. Veloso prepared the introduction and results. Yutuc prepared the discussion and conclusion.

In conclusion, logistic regression yielded unsatisfactory performance results, whereas random forest demonstrated satisfactory results, likely due to the quality of the dataset. This shows the flexibility of Random Forest in comparison to Logistic Regression. Aside from the models, the results show that the NCR (44%) has a high potential for an outbreak, while Region V (0%) has a low potential for an outbreak, with an average of 31.1% for all the regions.

References

Carvajal, Thaddeus M., et al. “Machine Learning Methods Reveal the Temporal Pattern of Dengue Incidence Using Meteorological Factors in Metropolitan Manila, Philippines.” *BMC Infectious Diseases, vol. 18*, no. 1, 17 Apr. 2018, www.ncbi.nlm.nih.gov/pmc/articles/PMC5905126/, https://doi.org/10.1186/s12879-018-3066-0.

Department of Health (2025). *EB Weekly Disease Surveillance Report.* https://doh.gov.ph

Philippine Statistics Authority (2023). *Health statistics report.* https://psa.gov.ph

Song, Jie, et al. “The Random Forest Model Has the Best Accuracy among the Four Pressure Ulcer Prediction Models Using Machine Learning Algorithms.” *Risk Management and Healthcare Policy, vol. Volume 14,* Mar. 2021, pp. 1175–1187, dx.doi.org/10.2147%2FRMHP.S297838, https://doi.org/10.2147/rmhp.s297838.

United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development.* https://sdgs.un.org/2030agenda

United Nations Development Programme (2023). *Human development report: Philippines.* https://hdr.undp.org

World Bank. (2024). *Philippines economic update: Health resilience.* https://www.worldbank.org/en/country/philippines

World Health Organization (2022). *World malaria report 2022.* https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2022